Introduction to Cloud Computing

Engineering Cloud Software Systems

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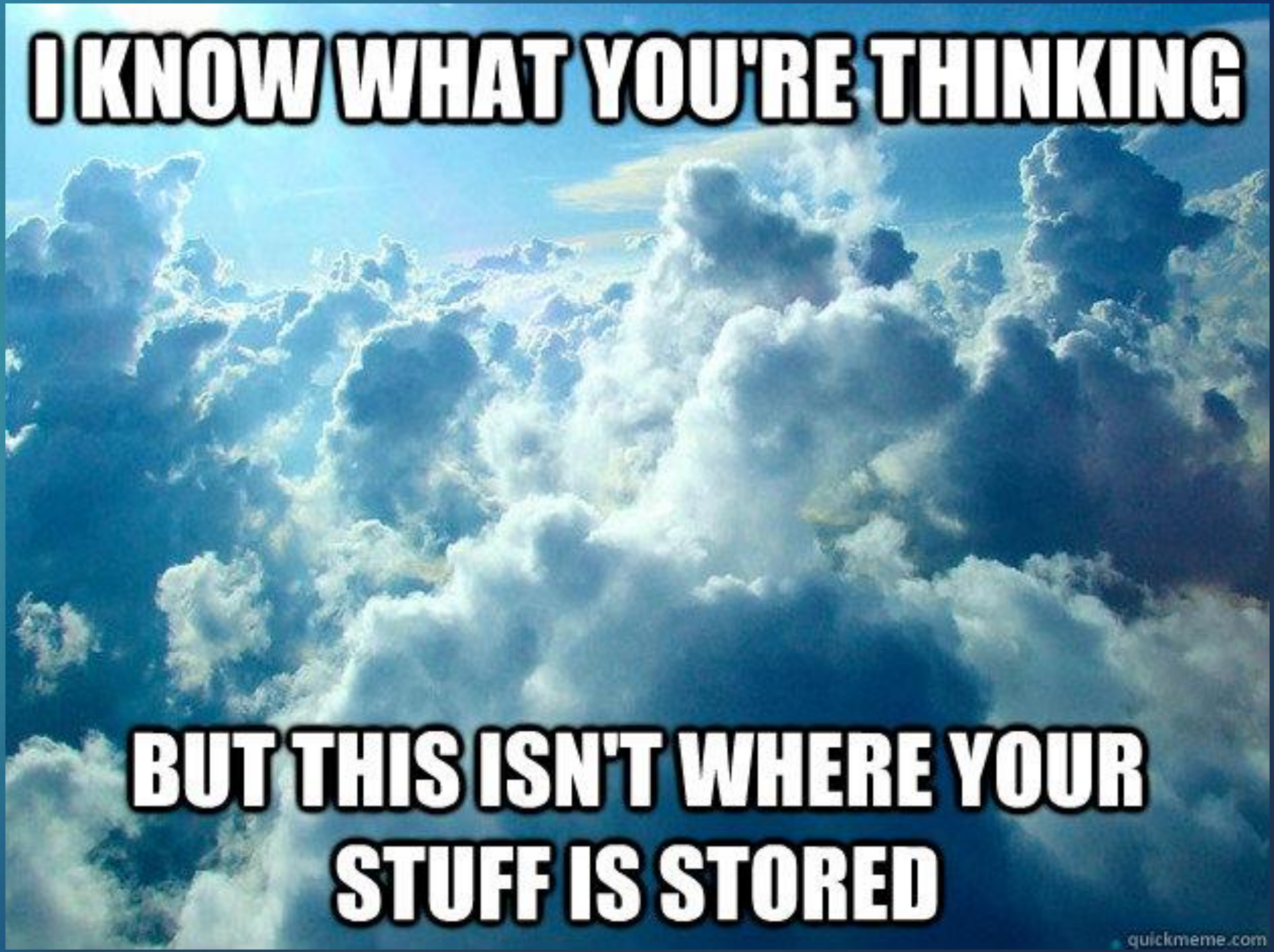
What is Cloud Computing?



Cloud computing means storing and accessing data and programs over the Internet instead of your computer's hard drive. The “cloud” is just a metaphor for the Internet.

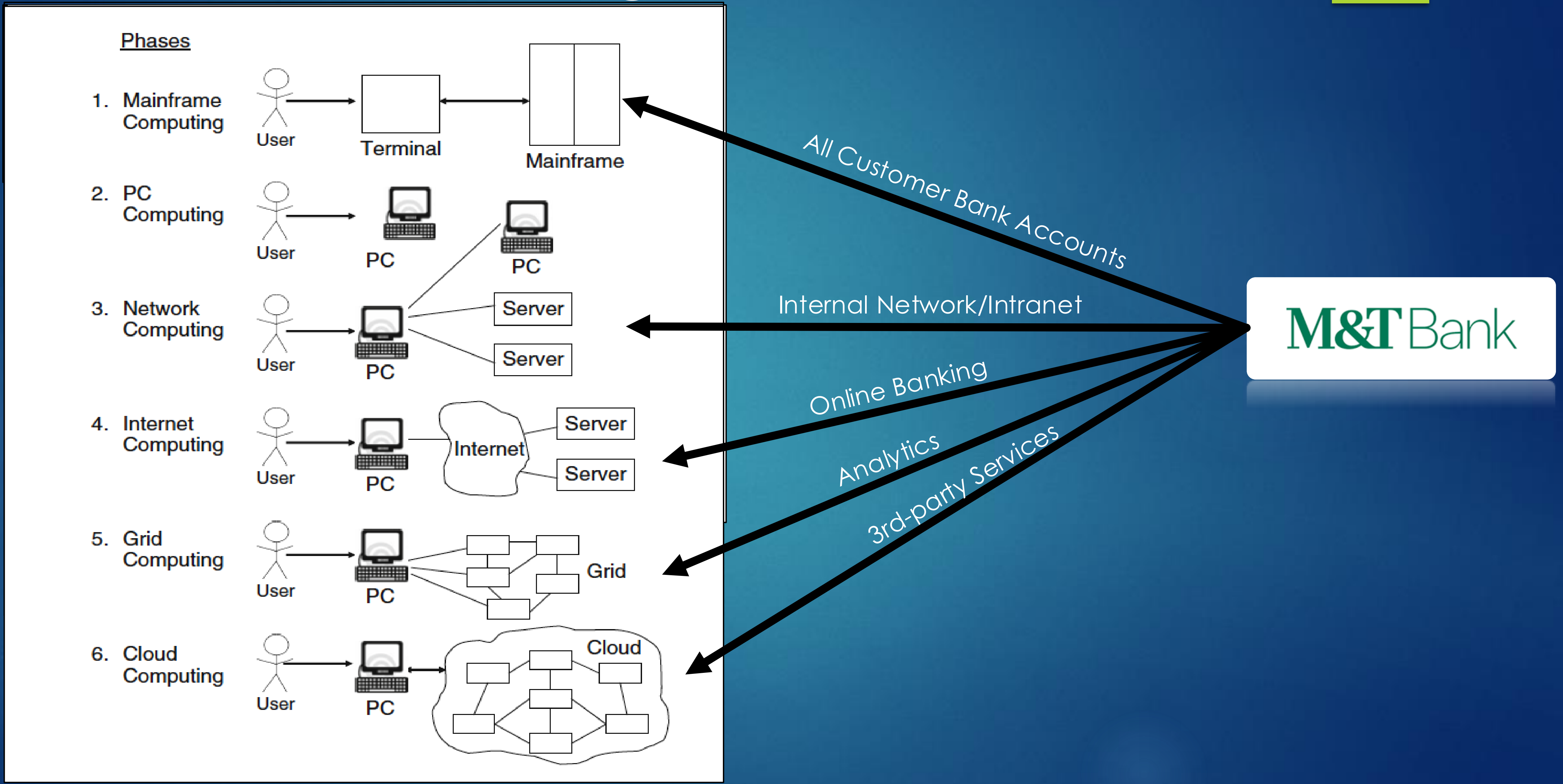
Source: https://www.pcmag.com/article/256563/what-is-cloud-computing

Simply put, cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change.

Source: https://azure.microsoft.com/en-us/overview/what-is-cloud-computing/


There is no cloud
it's just someone else's computer

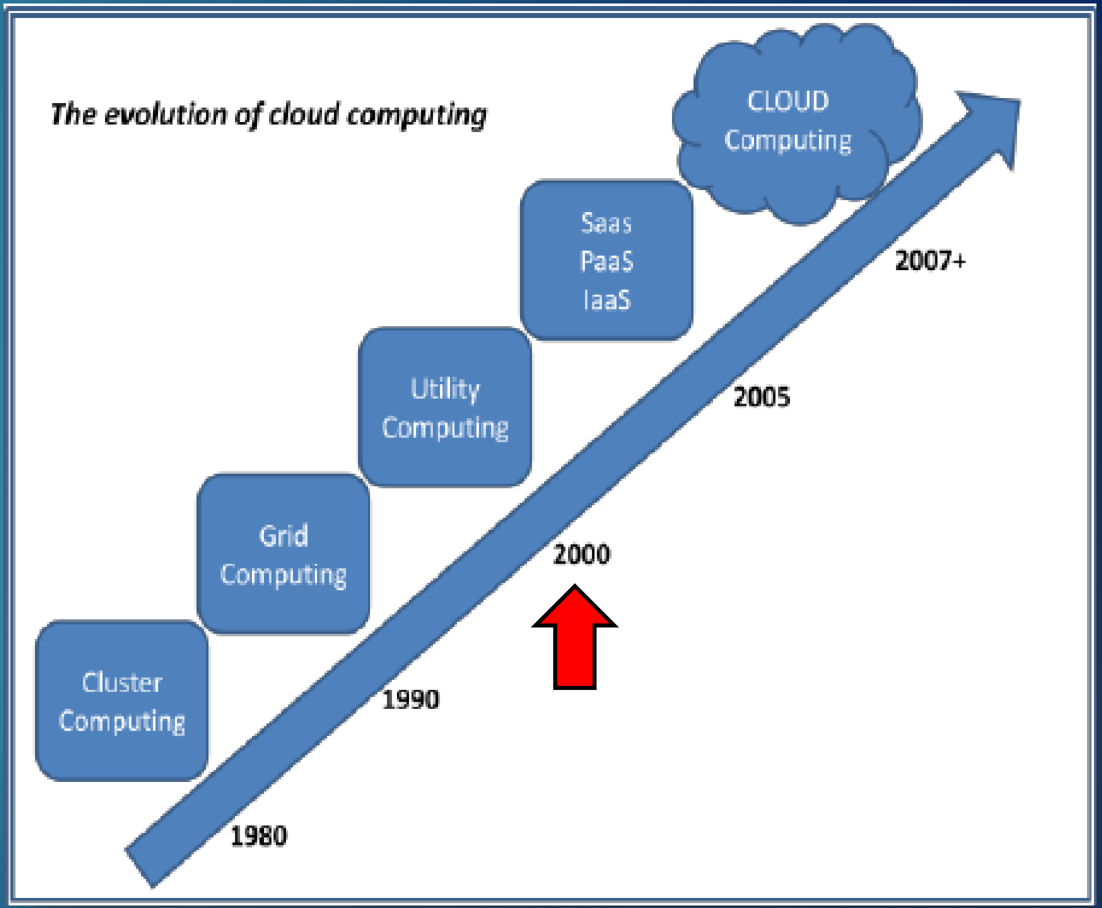
Phases of computing



Source: Voas, J., & Zhang, J. (March/April 2009). Cloud computing: New wine or just a new bottle? IEEE ITPro, 15–17

Key Dates in Cloud Computing

- When did Cloud Computing start?
- In existence since 2000
  - August 2006: Amazon created subsidiary Amazon Web Services (AWS)
  - April 2008: Google released Google App Engine (beta)
  - February 2010: Microsoft released Microsoft Azure



Key Principles of Cloud Computing

- Pooled Resources
- Virtualization
- Elasticity
- Automation
- Metered Billing

Pooled Resources

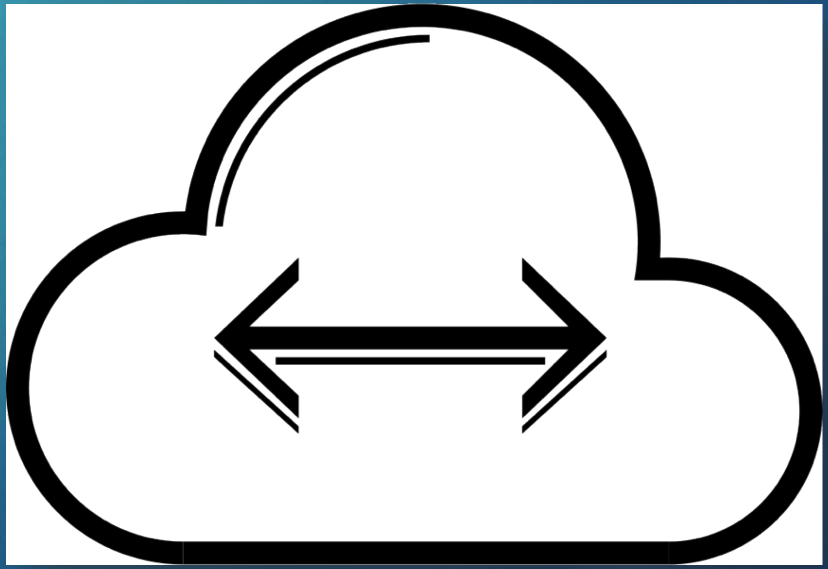
- The cloud provider's (e.g. AWS) computing resources are pooled to serve multiple consumers
- Different physical and virtual resources dynamically assigned and reassigned according to consumer demand
- More cost efficient as users subscribe to resources vs. purchasing
- Consumers share the same resources, which is referred to as "multi-tenancy"
  - This can have both advantages and disadvantages

Virtualization

- Vital to the cloud as physical servers consume significant space for power and cooling
- Each physical server is partitioned into many virtual servers serving multiple customers
- They constitute a large pool of resources available when required

Elasticity

- A typical application may require a base level of resources under normal conditions, but need more resource under peak load conditions
- Elasticity refers to the ability to dynamically change how much resource is consumed in response to how much is needed
- Without elasticity, a company would need to purchase hardware for their peak performance
  - Why is this a bad thing?



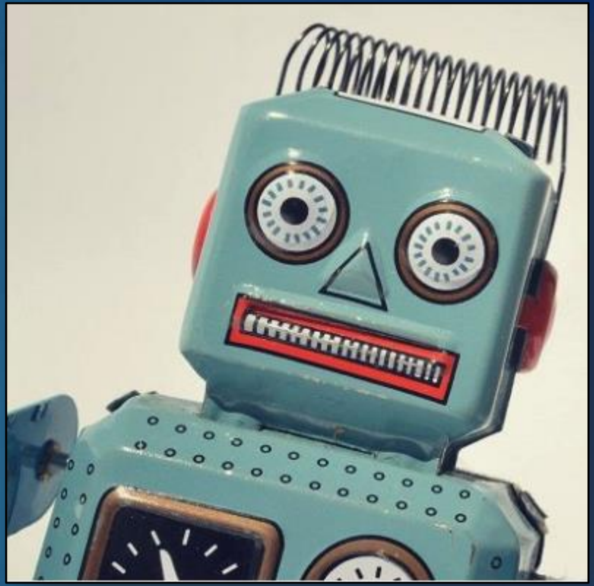
Anyone pre-order a Switch on April 24, 2025?

- Nintendo Switch 2 pre-orders crashed the several websites
- Over 2,000 shoppers were unable to access Target's website for about 45 minutes after pre-orders went live
- Best Buy experienced outages for several hours on the same day, with 56% of its roughly 600 reports citing login failures.
- Walmart's app also went down for about 15 minutes around before the pre-order were opened, with 55% of 1,700+ complaints reporting access issues
- Could elasticity have helped with the surge in usage?
  - Maybe, though not always easy as you'd think

Source: https://timesofindia.indiatimes.com/technology/gaming/nintendo-switch-2-pre-orders-crash-websites-of-multiple-retailers-in-us/articleshow/120595382.cms

Automation

- The ability to automatically provision and deploy a new instance of a machine and free or de-provision an instance when not needed
- Cloud-deployed application can provision new instances on an as-needed basis and brought online within minutes
- After the peak demand subsides and you don’t need the additional resources, virtual instances can be taken offline and de-provisioned, so that a customer will not be billed

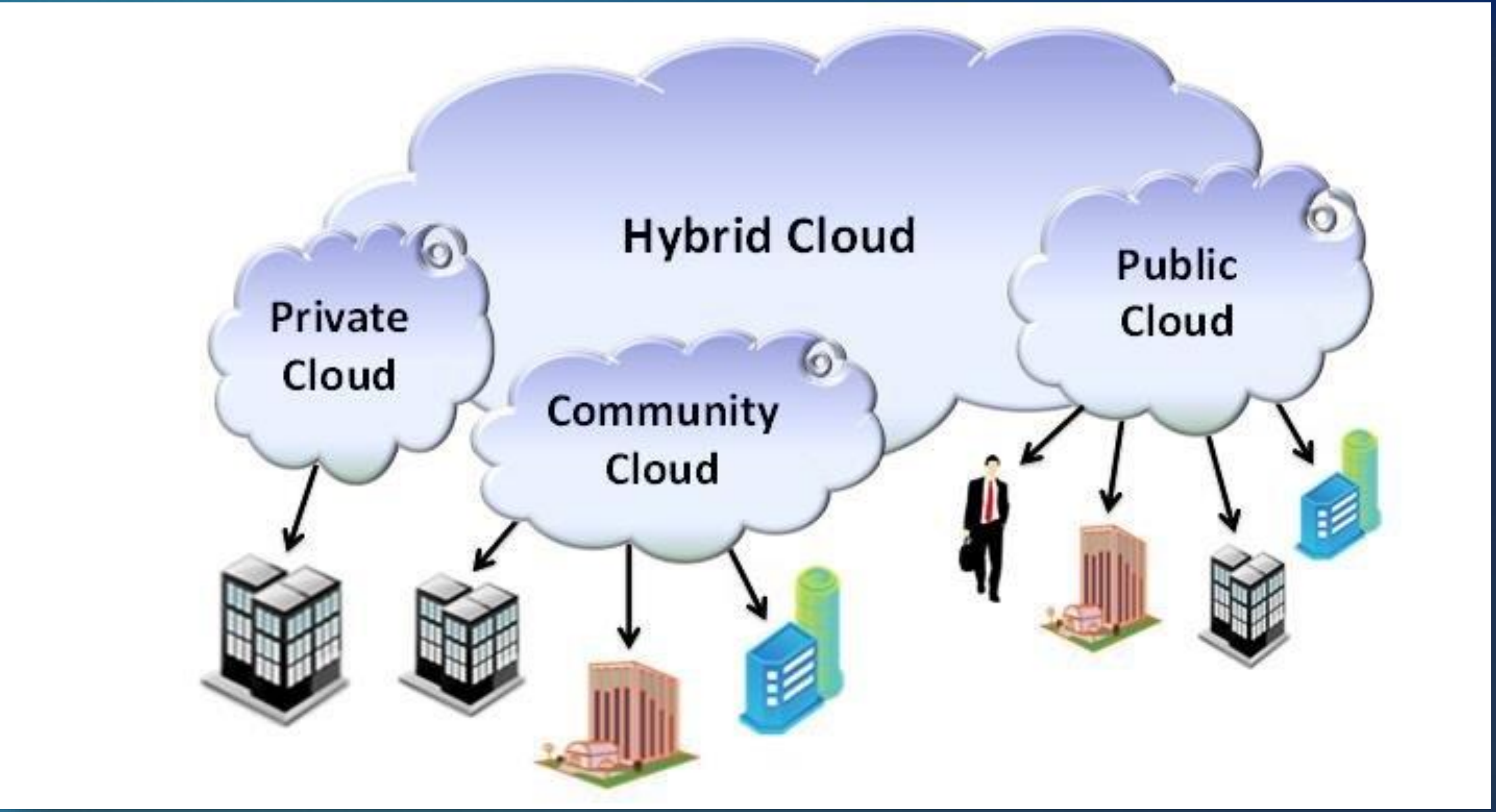


Metered Billing

- You only pay for what you use (mirrors the billing model of an electrical utility grid)
- No annual contract and no commitment for a specific level of consumption
- Completely changes the game for new start-ups that require lots of infrastructure
    - Previously all this had to be purchased ahead a time
- If you created a server on the cloud but it's not doing anything, are you paying anything for it?
- Important Lesson!
    - If you are not using a resource, turn it <u>OFF</u>

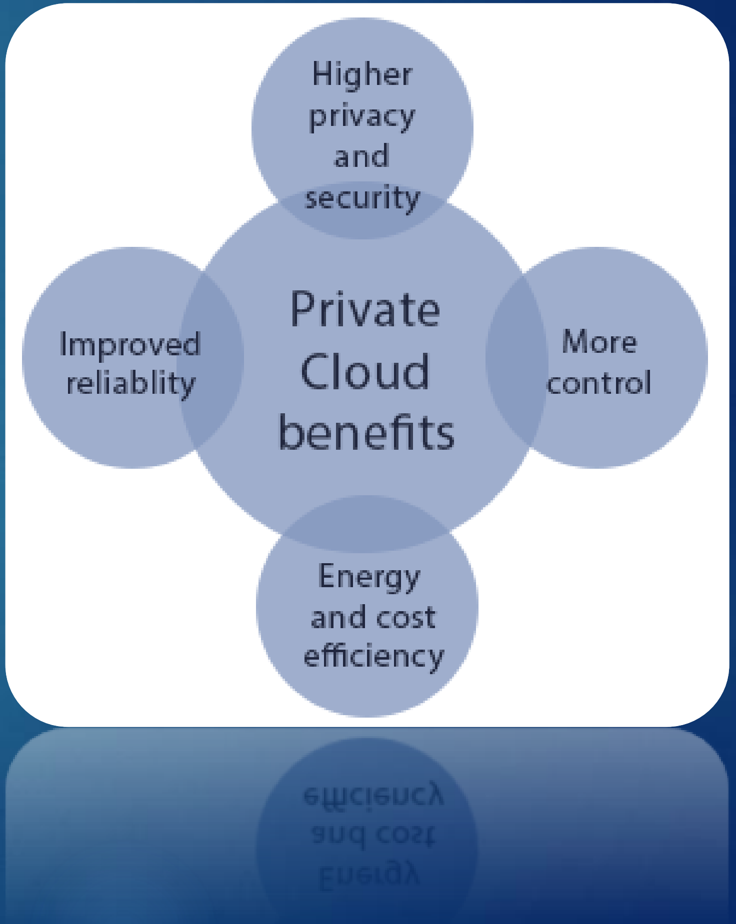
Cloud Deployment Models

1. Private Cloud
2. Public Cloud
3. Community Cloud
4. Hybrid Cloud



Private Cloud

- Owned, managed and operated by organization
- Can be onsite or offsite
- Usually selected when external mandates such as regulations and legislative requirements require a high degree of access accountability, control, and governance
- Examples: Banking, Healthcare, Gaming, Education



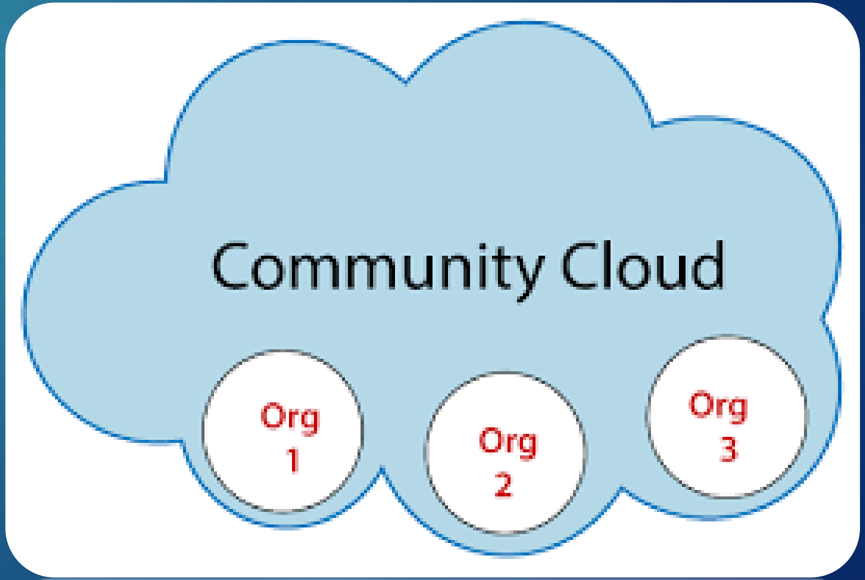
Source: https://www.rivacrmintegration.com/2016/private-cloud/

Public Cloud

- Cloud infrastructure resources available to the general public
- Offers efficiencies of scale, performance and resiliency that would be cost prohibitive to organizations doing this on their own
- Examples: AWS, Microsoft Azure and Google Cloud

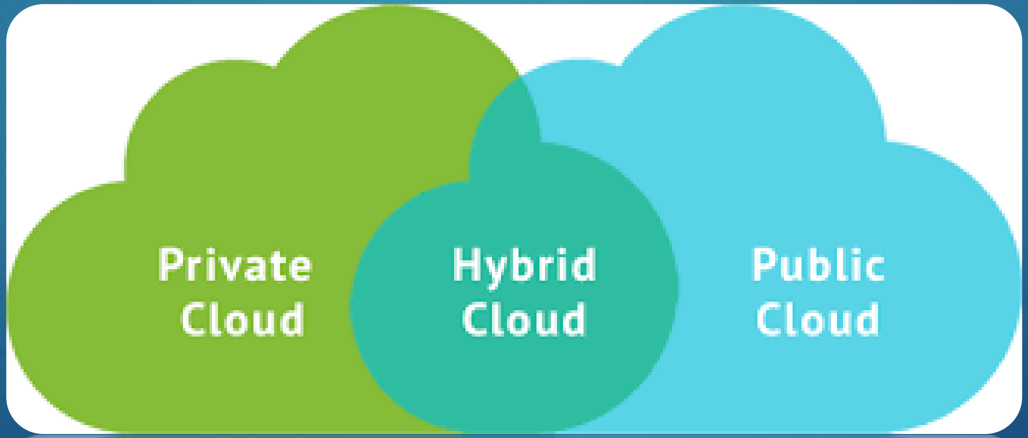
Community Cloud

- Shared infrastructure among multiple organizations, including servers, storage, and networking resources
- Enhanced security measures to protect shared infrastructure and data
- Designed to meet specific industry regulations and compliance standards
- Partitioned public clouds with isolated services for specific groups
  - AWS GovCloud
- Other examples: Healthcare, Financial



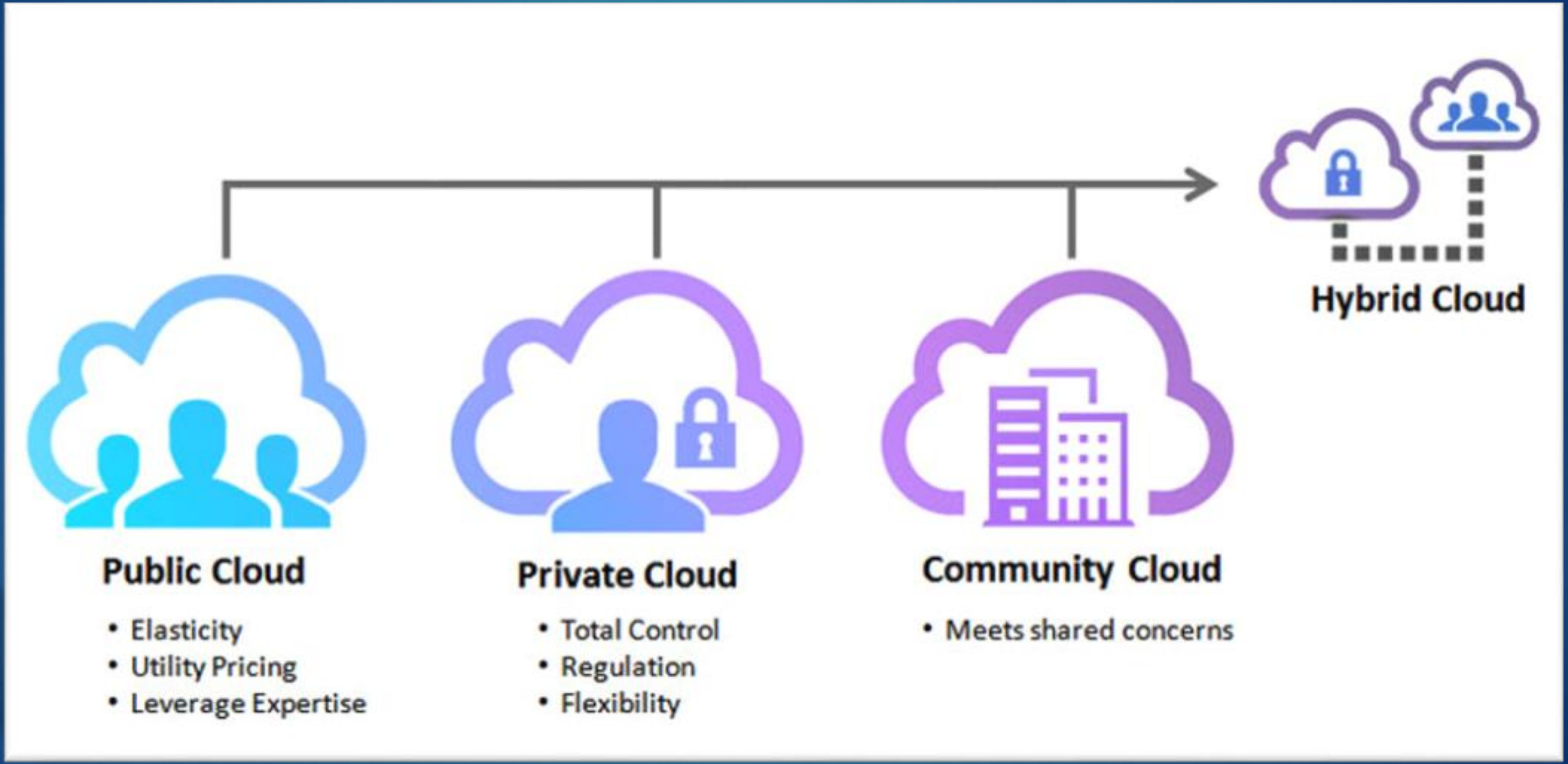
Hybrid Cloud

- The infrastructure is a composition of private, public or even community clouds
- They are connected to provide additional capacity or other computing resources
- Can provide the ability to support "cloud bursting" where additional capacity is needed that's not available on private cloud



Source: https://www.hyve.com/cloud-hosting/hybrid-cloud/

Deployment Models - Summary



Source: https://apextechno.co.uk/services/hybrid-cloud

- Look for a Slack workspace invite